Statistical Analysis of Road Accident Data

1. Introduction

Statistical analysis helps summarize large accident datasets into meaningful information. Using descriptive statistics, trends, distributions, and comparative measures, stakeholders can identify accident-prone areas and improve road safety planning.

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2. Descriptive Statistics

Statistic	Description	Formula
Mean	Average number of casualties	ΣX / N
Median	Middle value after arranging observations	Middle Observation
Mode	Most frequently occurring value	Highest Frequency
Minimum	Smallest casualty value	MIN(X)
Maximum	Largest casualty value	MAX(X)
Range	Difference between maximum and minimum	Max − Min
Standard Deviation	Spread of observations	σ
Variance	Square of standard deviation	σ^2

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3. Mean (Average)

The mean represents the average number of casualties recorded per accident.

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

where:

- **ΣX** = Total casualties
- **N** = Total accidents

Dashboard Calculation

Total Casualties = **195,737**

Total Accidents = **144,419**

Mean

$$= \frac{195737}{144419} = 1.36$$

Interpretation

On average, **1.36 casualties** occurred in every reported road accident.

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4. Median

The median represents the middle observation after arranging all casualty values in ascending order.

Formula

If **n** is odd

$$Median = \frac{n+1}{2}$$

If **n** is even

$$Median = \frac{n/2 + (n/2 + 1)}{2}$$

Interpretation

The median is less affected by extreme accident cases than the mean and provides a better measure of central tendency when casualty data is skewed.

Note: The exact median value requires the raw Number_of_Casualties data and cannot be calculated from dashboard summaries alone.

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5. Mode

The mode is the casualty value that occurs most frequently.

Formula

Mode = Observation with Highest Frequency

Example

Casualties	Frequency
1	55,000
2	42,000
3	18,000

Mode = **1 Casualty**

Interpretation

The mode identifies the most common accident severity in terms of casualties.

Note: The exact mode requires the raw dataset and cannot be computed from summary totals alone.

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6. Minimum and Maximum

Minimum indicates the smallest casualty count recorded.

Maximum indicates the largest casualty count recorded.

These values help identify the range of accident severity.

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7. Range

$$Range = Maximum - Minimum$$

A larger range indicates greater variation in casualty counts.

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8. Standard Deviation

Standard deviation measures the spread of casualty values around the mean.

- Small value → casualties are consistent.
- Large value → casualty counts vary significantly.

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9. Variance

Variance measures the dispersion of casualty values.

$$Variance = (Standard\ Deviation)^2$$

Higher variance indicates greater inconsistency in accident severity.

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10. Accident Severity Analysis

Severity	Casualties	Percentage
Fatal	2.9K	1.5%
Serious	27.0K	13.8%
Slight	165.8K	84.7%

Interpretation

Most accidents resulted in **slight injuries**, while fatal accidents accounted for a very small proportion of the total casualties.

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11. Area Analysis

Area	Percentage
Urban	61.9%
Rural	38.1%

Urban areas experienced a higher proportion of casualties, likely due to greater traffic density.

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12. Light Condition Analysis

Condition	Percentage
Daylight	73.8%
Darkness	26.2%

The majority of casualties occurred during daylight hours, reflecting higher daytime traffic volumes.

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13. Year-over-Year Analysis

Year	Casualties
2021	Higher
2022	195.7K

Overall Change:

−11.9%

This indicates an overall reduction in casualties compared with the previous year.

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14. Business Insights

- Average casualties per accident are approximately **1.36**, indicating that most accidents involve one or two individuals.
- **Slight injuries** account for the majority of casualties, while fatal accidents are comparatively rare.
- **Urban areas** contribute the largest share of casualties due to higher traffic density.
- **Single carriageways** are associated with the highest number of casualties, suggesting they should be prioritized for safety improvements.
- The **11.9% year-over-year reduction** indicates progress in road safety, though continued interventions remain necessary.

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15. Conclusion

The statistical analysis demonstrates how descriptive statistics support evidence-based road safety decisions. While dashboard metrics provide valuable high-level insights, measures such as the **exact median, mode, minimum, maximum, variance, and standard deviation** require access to the raw accident records. Integrating these measures with interactive Power BI visualizations enables authorities to identify trends, prioritize high-risk locations, and implement targeted safety initiatives.